ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording. Symbols and circuits may be drawn in different but correct ways.

QUESTION 1

1.1 Mechatronics is the combination of mechanical engineering, electronics / electrical engineering and computer control (programming) ✓ to design and build intelligent machines and products ✓. (2)

1.2.1 First-class lever (the fulcrum is between the effort and the load) ✓ (1)

1.2.2 $\text{Moments about the fulcrum: load} \times 0{,}3 = 50 \times 1{,}2$ ✓; $\text{load} = 60 \div 0{,}3$ ✓ $= 200\ \text{N}$ ✓ (3)

1.2.3 $\text{MA} = \text{load} \div \text{effort} = 200 \div 50$ ✓ $= 4$ ✓ (2)

1.3.1 $N_2 = N_1 \times D_1 \div D_2$ ✓ $= 1\,440 \times 100 \div 300$ ✓ $= 480\ \text{r/min}$ ✓ (3)

1.3.2 $\text{VR} = D_2 \div D_1 = 300 \div 100$ ✓ $= 3\ (3:1)$ ✓ (2)

1.4 Any TWO, ✓✓ each: no slip, so the speed ratio stays exact; can transmit larger torque / power; compact (short centre distance); long life and little stretching. (4)

1.5.1 Light (solar) energy to electrical energy ✓

1.5.2 Electrical energy to sound energy ✓

1.5.3 Electrical energy to heat energy ✓

(3)

[20]

QUESTION 2

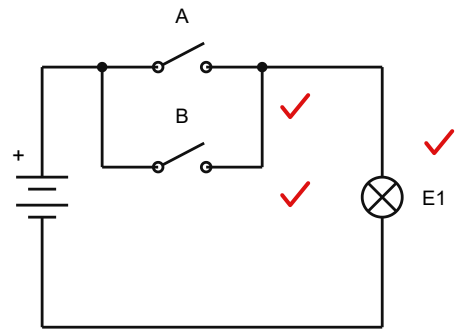
2.1 The current through a conductor is directly proportional to the voltage across it ✓, provided the temperature (resistance) stays constant ✓ (V = I × R). (2)

2.2.1 $R_T = R_1 + R_2 + R_3 = 10 + 20 + 30$ ✓ $= 60\ \Omega$ ✓ (2)

2.2.2 $I = V \div R_T = 12 \div 60$ ✓ $= 0{,}2\ \text{A}$ ✓ (2)

2.2.3 $V_3 = I \times R_3 = 0{,}2 \times 30$ ✓ $= 6\ \text{V}$ ✓ (2)

2.3



A	B	E1
0	0	0
0	1	1
1	0	1
1	1	1

Switches A and B in parallel ✓✓, in series with lamp E1 across the supply ✓; truth table: E1 = 1 when A OR B (or both) is 1 ✓✓✓

(6)

2.4



2.4.1 fuse 2.4.2 NC contact

2.4.1 Fuse: rectangle with the line through it ✓✓ 2.4.2 NC contact: closed contact ✓✓

(4)

2.5.1 Light-emitting diode ✓

2.5.2 Alternating current ✓

(2)

[20]

QUESTION 3

3.1 It stores compressed air ✓ so that air is available when demand is high; it smooths out pressure pulses from the compressor ✓; it lets the air cool so that water condenses and can be drained ✓. (3)

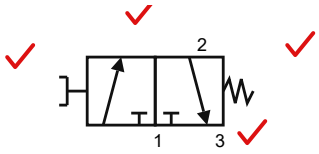
3.2.1 1.0 – double-acting cylinder ✓; 1.1 – 5/2-way valve, lever operated with spring return ✓ (2)

3.2.2 The valve changes over ✓; air flows from 1 to 4 into the piston side of the cylinder ✓ while the rod side exhausts through 2 to 3, so the cylinder extends ✓. (3)

3.2.3 The spring returns the valve to its rest position ✓; air flows from 1 to 2 into the rod side and the cylinder retracts ✓. (2)

3.2.4 A one-way flow control valve ✓ fitted in the exhaust (rod-side) line, meter-out ✓ (2)

3.3



Two squares ✓, ports 1, 2 and 3 correctly shown with 1 blocked at rest ✓, push-button actuator ✓, spring return ✓

(4)

3.4 $A = \pi \times d^2 \div 4 = \pi \times 0{,}05^2 \div 4$ ✓ $= 1{,}963 \times 10^{-3}\ m^2$ ✓; $F = p \times A = 500\ 000 \times 1{,}963 \times 10^{-3}$ ✓ $= 981{,}7\ N$ ✓ (4)

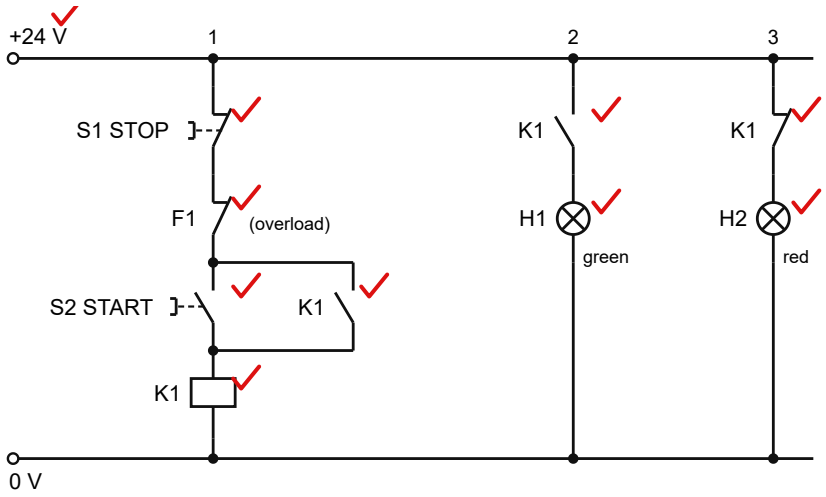
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QUESTION 4

4.1 A contactor is an electrically operated switch ✓ that lets a small control current switch the large current of a motor (main circuit) on and off ✓. (2)

4.2 An isolator is a manual switch used to disconnect the supply safely for maintenance and must not be opened under load ✓✓. A circuit breaker switches off automatically when there is an overload or short circuit and can be reset ✓✓. (4)

4.3



Supply lines ✓; STOP S1 (NC) ✓ and overload F1 (NC) ✓ in series; START S2 (NO) ✓ with a K1 holding contact in parallel ✓; coil K1 ✓; K1 NO contact ✓ to green lamp H1 ✓; K1 NC contact ✓ to red lamp H2 ✓

(10)

4.4 It switches the motor off when it draws too much current for too long ✓, so that the motor windings do not overheat and burn out ✓. (2)

4.5 Red button ✓ on a yellow background ✓ (2)

[20]

QUESTION 5

5.1 Any FOUR ✓✓✓✓: wear the correct PPE; no loose clothing or jewellery near machines; keep the floor clean and dry; switch off and isolate machines before working on them; use tools only for their purpose; report all accidents and faults; know where the emergency stops and fire extinguishers are. (4)

5.2.1 Safety goggles / face shield ✓

5.2.2 Leather gloves ✓

5.2.3 Ear plugs / ear muffs ✓

5.2.4 Safety glasses (and fume extraction) ✓

(4)

5.3 Any FIVE in a logical order ✓✓✓✓✓: check that the supply is on and the isolator is closed; check the fuses / circuit breaker; check that the emergency stop is released and the overload is not tripped; check the START and STOP buttons and the control wiring; check whether contactor K1 pulls in; check the motor connections and the motor itself; check that the conveyor is not jammed. (5)

5.4 A routine inspection is a regular, quick check of the condition of a machine (e.g. listening, looking for leaks, checking oil levels) ✓✓. An overhaul is a major service in which the machine is stripped, worn parts are replaced and it is rebuilt ✓✓. (4)

5.5 Any THREE ✓✓✓: to see which faults repeat; to plan preventive maintenance; to know which spares to keep in stock; to calculate downtime and repair costs; as a legal / safety record. (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE